

MATH 526 Lab 1 Fall 2006

Part I

This is an introduction to MATLAB. First you should figure out how to start a MATLAB command window. (You may want to create a shortcut for future convenience.) Try the following commands inside the MATLAB command window. (The commands are in boldfaced letters.)

```
>> help help
```

The command **help** is very useful!

Try some elementary calculations:

```
>> 2 + 3
```

```
>> 4 + sqrt(2)
```

```
>> 5 + sqrt(2);
```

The semicolon suppress the answer. To see it, type

```
>> ans
```

We input a row vector as:

```
>> v = [1 2 3] (or v=[1,2,3])
```

and a column vector as:

```
>> w = [1; 2; 3]
```

Is there a difference if we type the following?

```
>> w = [1  
      2  
      3]
```

We input a matrix as follows:

```
>> A = [1 2 3; 4 5 6; 7 8 9] (or A = [1,2,3;4,5,6;7,8,9])
```

or as

```
>> A = [1 2 3  
      4 5 6  
      7 8 9]
```

We can easily change an entry in a vector or matrix. Try:

```
>> w(1) = 0
```

```
>> A(2,3) = 10
```

The transpose of a vector or matrix is obtained as follows.

```
>> w'
```

```
>> A'
```

Suppose we forgot what variables have been defined. We can retrieve the information using the commands:

```
>> who
```

```
>> whos
```

What is the difference?

If you want to see what a variable represents, just type the name of the variable. For example you can view the matrix **A** and the vectors **v**, **w** again by typing

```
>> A, v, w
```

We now try some operations on vectors and matrices. Let

```
>> B = [1 0 -1; 0 3 5; 6 2 3]
```

```
>> C = [4 -1 3; 0 2 -5]
```

Perform the following operations:

```
>> z = v + 2*w'
```

```
>> A + B
```

```
>> A - B
```

```
>> A*B
```

```
>> C*A
```

```
>> A*C
```

What happened?

```
>> (A*B)'
```

```
>> A' * B'
```

```
>> B' * A'
```

Which one of **A'*B'** and **B'*A'** equals **(A*B)'**?

We can make a larger matrix from smaller parts as follows:

```
>> D = [A B]
```

```
>> E = [A B  
       B A]
```

You can also create **E** by the command **E = [A B; B A]**.

We can pick out parts of a matrix. What do the following commands do? (Consult Section 11 of the MATLAB Primer.)

```
>> D(:,3)
```

```
>> D(2,:)
```

```
>> E(1:3, :)
```

```
>> E(2:4, 1:3)
```

Some special matrices are built into MATLAB. Try

```
>> help ones
```

```
>> help zeros
```

```
>> help eye
```

and create the 3×3 zero matrix, the 4×4 identity matrix, and the 5×5 matrix with all entries equal to 2.

To quit matlab, type

```
>> quit
```

Restart MATLAB and proceed to Part II.